

Expiement 8

```
!pip install transformers datasets evaluate sentencepiece sacrebleu rouge-score
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Collecting evaluate
  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Collecting sacrebleu
  Downloading sacrebleu-2.6.0-py3-none-any.whl.metadata (39 kB)
Collecting rouge-score
  Downloading rouge_score-0.1.2.tar.gz (17 kB)
  Preparing metadata (setup.py) ... done
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2024.11.23)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.20.1)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.5.1)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.17)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (2024.11.0)
Collecting portalocker (from sacrebleu)
  Downloading portalocker-3.2.0-py3-none-any.whl.metadata (8.7 kB)
Requirement already satisfied: tabulate>=0.8.9 in /usr/local/lib/python3.12/dist-packages (from sacrebleu) (0.9.0)
Collecting colorama (from sacrebleu)
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: lxml in /usr/local/lib/python3.12/dist-packages (from sacrebleu) (6.0.2)
Requirement already satisfied: absl-py in /usr/local/lib/python3.12/dist-packages (from rouge-score) (1.4.0)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (from rouge-score) (3.9.1)
Requirement already satisfied: six>=1.14.0 in /usr/local/lib/python3.12/dist-packages (from rouge-score) (1.17.0)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[h]) (4.0.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.27.0)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.24.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub) (4.12.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2) (3.3.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2) (2025.1.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk->rouge-score) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk->rouge-score) (1.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (2.5.0)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (6.3.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (0.2.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (1.18.0)
```

```
import torch
import pandas as pd
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
from datasets import Dataset
from sklearn.model_selection import train_test_split
from evaluate import load
```

```
df = pd.read_csv("bank_faq.csv")
```

```
print(df.head()) # check data
```

	question	answer
0	What is ATM?	ATM is a machine used to withdraw cash from ba...
1	What is net banking?	Net banking allows customers to perform bankin...
2	What is mobile banking?	Mobile banking allows banking services through...

```

3     What is IFSC code? IFSC code identifies a specific bank branch fo...
4     What is a debit card? A debit card allows you to access money from y...

```

```

train_q, test_q, train_a, test_a = train_test_split(
    df["question"], df["answer"], test_size=0.3, random_state=42
)

```

```

train_dataset = Dataset.from_dict({
    "question": list(train_q),
    "answer": list(train_a)
})

```

```

test_dataset = Dataset.from_dict({
    "question": list(test_q),
    "answer": list(test_a)
})

```

```
model_name = "google/mt5-small"
```

```

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:

The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/token>)

You will be able to reuse this secret in all of your notebooks.

Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
```

```
config.json: 100%                               553/553 [00:00<00:00, 11.8kB/s]
```

```
tokenizer_config.json: 100%                     82.0/82.0 [00:00<00:00, 1.67kB/s]
```

```
spiece.model: 100%                             4.31M/4.31M [00:00<00:00, 33.3MB/s]
```

```
special_tokens_map.json: 100%                  99.0/99.0 [00:00<00:00, 3.42kB/s]
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits a

WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_

```
pytorch_model.bin: 100%                       1.20G/1.20G [00:06<00:00, 262MB/s]
```

```
model.safetensors: 100%                      1.20G/1.20G [00:16<00:00, 59.8MB/s]
```

```
Loading weights: 100%                        192/192 [00:00<00:00, 718.15it/s, Materializing param=shared.weight]
```

The tied weights mapping and config for this model specifies to tie shared.weight to encoder.embed_tokens.weight, but

The tied weights mapping and config for this model specifies to tie shared.weight to decoder.embed_tokens.weight, but

The tied weights mapping and config for this model specifies to tie shared.weight to lm_head.weight, but both are pre

```
generation_config.json: 100%                   147/147 [00:00<00:00, 3.63kB/s]
```

```

def preprocess(batch):
    inputs = tokenizer(batch["question"], padding="max_length", truncation=True, max_length=64)
    targets = tokenizer(batch["answer"], padding="max_length", truncation=True, max_length=64)

    inputs["labels"] = targets["input_ids"]
    return inputs

```

```

train_data = train_dataset.map(preprocess, batched=True)
test_data = test_dataset.map(preprocess, batched=True)

```

```
Map: 100%                                     38/38 [00:00<00:00, 642.57 examples/s]
```

```
Map: 100%                                     17/17 [00:00<00:00, 500.24 examples/s]
```

```
optimizer = torch.optim.AdamW(model.parameters(), lr=3e-5)
```

```
model.train()
```

```

for epoch in range(5):
    total_loss = 0

```

```

    for batch in train_data:
        optimizer.zero_grad()

```

```
        input_ids = torch.tensor(batch["input_ids"]).unsqueeze(0)
```

```
        labels = torch.tensor(batch["labels"]).unsqueeze(0)
```

```

        outputs = model(input_ids=input_ids, labels=labels)
        loss = outputs.loss

        loss.backward()
        optimizer.step()

        total_loss += loss.item()

    print(f"Epoch {epoch+1} Loss:", total_loss)

```

```

Epoch 1 Loss: 1434.981008529663
Epoch 2 Loss: 1274.0837173461914
Epoch 3 Loss: 1133.052682876587
Epoch 4 Loss: 1091.4537181854248
Epoch 5 Loss: 1039.246810913086

```

```

model.eval()

predictions = []
references = []

for batch in test_data:
    input_ids = torch.tensor(batch["input_ids"]).unsqueeze(0)

    outputs = model.generate(input_ids)
    pred = tokenizer.decode(outputs[0], skip_special_tokens=True)

    ref = tokenizer.decode(batch["labels"], skip_special_tokens=True)

    predictions.append(pred)
    references.append(ref)

```

```

rouge = load("rouge")
bleu = load("bleu")

rouge_score = rouge.compute(predictions=predictions, references=references)
bleu_score = bleu.compute(
    predictions=predictions,
    references=[[r] for r in references]
)

print("ROUGE-L:", rouge_score["rougeL"])
print("BLEU:", bleu_score["bleu"])

```

```

ROUGE-L: 0.00980392156862745
BLEU: 0.0

```

```

def chatbot(question):
    inputs = tokenizer(question, return_tensors="pt")
    outputs = model.generate(**inputs)
    return tokenizer.decode(outputs[0], skip_special_tokens=True)

print(chatbot("why atm used?"))

```

```
<extra_id_0>
```

Start coding or [generate](#) with AI.

